User Interface



Q. Is this tomato ripe?

Yes (Ripe) | No (Not Ripe)

Human operator

Robot & Environment



y

q^*

o_t

a_t

Planning System

Knowledge Base (KB)

- facts about objects
- robot states / actions
- task knowledge

Planner

Belief over Possible Worlds





$s_1: 0.6$ $s_2: 0.3$ ... $s_N: 0.1$

Belief Update

$conf > \tau$

Select Query

Update KB